Third Semester B. Tech. (Electronics and Communication Engineering) Examination

ELECTRONIC CIRCUITS

Time : 3 Hours] [Max. Marks : 50

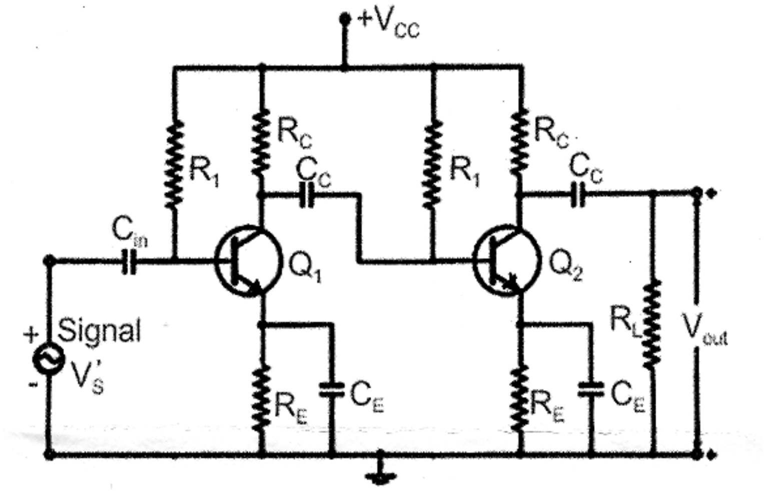
Instructions to Candidates :—

(1) All questions carry marks as indicated.
(2) Illustrate your answers with a neat digram wherever necessary.
(3) Internal choice is available in Question No. **Two**.

1. (A) In a class B push-pull amplifiers how would you justify connecting two diodes (or resistances) in the base circuit results in elimination of crossover distortion ? 4(CO4)

   (B) A power transistor operated in class A operation delivers a maximum of 5 W to $R_L = 4\,\Omega$ load. Assume supply voltage of 20 V. The quiescent operating point is adjusted for a symmetrical swing.
   Estimate :
   (i) Transformer turns ratio.
   (ii) Maximum collector current.
   (iii) Co-ordinates of quiescent operating point.
   (iv) Conversion efficiency. 6(CO4)

2. For a two stage RC coupled amplifier, the circuit component values are: $R_{b1} = R_{b2} = 1\,M\Omega$; $R_{c1} = R_{c2} = 5K\,\Omega$; $R_{c1} = R_{c2} = 1K\,\Omega$; $RL = 10\,K\Omega$.
   Assume $h_{fe} = \beta = 100$ and $h_{ie} = 2\,K\Omega$. Construct its low frequency model and estimate values of voltage gain of individual stages and overall voltage gain.
   [Fig On Next page]



10(CO3)

OR

2. (A) Explain the frequency response of RC Coupled transistor amplifier. 5(CO3)

   (B) In which situation do we use Darlington Pair ? Draw and illustrate your answer. 5(CO3)

3. (A) An electronic manufacturing unit has designed an amplifier with an open loop gain Av = 100. In order to improve amplifier performance negative feedback is deployed that results in reduction of overall gain to 50.

   (a) Can you estimate feedback factor β ?

   (b) If the calculated value of β(in part a) is maintained, estimate amplifier gain, if the overall stage gain is 75. 5(CO1)

   (B) A Hartley oscillator is employed in a T.V. receiver circuit to generate sustained oscillations of frequencies in the range 950 KHz to 2050 KHz. The oscillator tank circuit is designed with $L_1 = 2$ mH and $L_2 = 20\ \mu$H and a variable capacitor. Compute range of capacitance values for generating sustained oscillations of frequencies in the range 950 KHz to 2050 KHz. 5(CO1)

4. (A) In an Astable multivibrator, for both transistors the resistance values are. $R_{b1} = R_{b2} = 5\ k\Omega$; $R_{c1} = R_{c2} = 0.4\ K\Omega$. The Coupling capacitors used are $C_1 = C_2 = 0.02\ \mu F$.

Estimate :

(i) Time period and frequency of circuit oscillations.

(ii) Minimum value of transistor β. 5(CO2)

(B) Analyze role of the timing components (R and C) and trigger pulse in Monostable Multivibrator and with a neat diagram, explain its operation. Also, mention the expression for time duration of output pulse. 5(CO2)

5. The problem statement for design of emitter follower type SVR is given below : Unregulated DC input is 12 V with source resistance of 2 Ω is available. Expected regulated output voltage is 5 V, with full load current of 200 mA.

Assume transistor has $h_{fe} = h_{FE} = 40$.

Design circuit, select components using appropriate design steps and determine its S_v and R_o. 10(CO5)